

Computer Vision Project

Submission deadline: 4/6/2026

Option 1 (preferred): Exercise on Transfer Learning

Weight	15 points	Group Size	Two students
Default Dataset	Oxford Flowers 102	Main Approach	Transfer learning

1. Project Overview

In this project, each group will develop an image-classification system using transfer learning and pretrained convolutional neural networks. The project will be run in a contest style in which all groups work on the same dataset and compare their final performance under a common evaluation setup.

2. Problem Statement

Build an image classifier for the Oxford Flowers 102 dataset. Given an input flower image, your system should predict the correct class label among 102 flower categories. You must use transfer learning and compare at least two different pretrained CNN backbones.

3. Dataset

The required dataset for this project is Oxford Flowers 102. You are responsible for obtaining the dataset yourself. You may obtain the dataset from either of the following sources:

- Official Oxford VGG dataset page: <https://www.robots.ox.ac.uk/~vgg/data/flowers/102/>
- TensorFlow Datasets (recommended): https://www.tensorflow.org/datasets/catalog/oxford_flowers102

Using TensorFlow Datasets is recommended because it simplifies downloading and preparing the dataset and provides ready-made train, validation, and test splits.

4. Required Technical Requirements

Each group must satisfy all of the following requirements:

1. Use transfer learning with pretrained CNN models. Training a deep model from scratch is not allowed as the main solution.
2. Use **two** pretrained backbones and compare them experimentally under a fair setup. The pretrained models to be used are MobileNetV2 and ResNet50V2 only.
3. Use the Oxford Flowers 102 dataset only. Do not use external labeled training data.
4. Report standard classification results, including at least accuracy, macro precision, macro recall, macro F1-score, and brief error analysis.
5. Select one final model as your best model and use it to generate your final predictions for the contest.

5. Contest Ranking

The contest ranking will be determined using the following criteria:

- Primary ranking metric: highest macro F1-score on the official test set.
- tie-breaker: higher classification accuracy.

6. Guidelines

The following guidelines are intended to keep the project feasible on modest compute resources such as Google Colab:

- Use reasonable image sizes such as 160×160, 192×192, or 224×224.
- Use data augmentation when appropriate and describe it clearly.
- You may experiment with frozen backbones, partial fine-tuning, learning rates, optimizers, and classifier-head design.
- Keep your experiments manageable and well documented.

For fairness, follow these rules:

- Do not use the test set to tune your model.
- Do not manually relabel or clean the dataset labels.
- Any preprocessing, augmentation, or sampling strategy must be documented.
- Your code and report must be your own work.

7. Deliverables

Each group must submit the following:

1. A report of about 4 to 6 pages that includes the problem definition, dataset source, preprocessing, models used, training procedure, experiments, results, confusion matrix, error analysis, and conclusions.
2. A short in-class 10-minute presentation.
3. All code used in the project, including report, presentation, training notebook(s) or scripts, evaluation code, and a short README file explaining how to run the project. Put all files in one compressed file and upload it to the project assignment on Team. **You do not need to include the trained model file.**

8. Suggested Workflow

1. Download and inspect the dataset.
2. Prepare the data pipeline and define a baseline model.
3. Train and evaluate your first pretrained backbone.
4. Train and evaluate a second pretrained backbone.
5. Improve your best model through controlled experiments.
6. Generate final predictions and summarize your findings in the report.

9. Grading

The project will be graded out of 15 points as follows:

Component	Points
Problem setup and data preparation	2
Implementation quality	2
Use and comparison of the two backbones	3
Evaluation and analysis	3
Contest performance	2

Report and presentation	3
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10. Important Notes

- Every group must use at least two pretrained backbones.
- Every group must clearly identify its best final model in the report.
- Final contest ranking is based on macro F1-score, then accuracy.
- You may need to use Google Colab or Kaggle to run your experiments in case you don't have a GPU.

Option 2: Suggest an Idea

A group may propose a different project idea instead of the default contest project, but it must be approved by the instructor in advance. Any proposed alternative must satisfy all of the following:

- it must be in computer vision,
- it could be using classical computer vision or include a deep learning component,
- it must be feasible with limited compute,
- it must include quantitative evaluation,
- and it must be comparable in difficulty to the default project.

The instructor has the right to reject or request modifications to any proposal.

The expected deliverables are similar to Option 1.

If you decide on this option, please send me a 1-page proposal by 21/5/2026.